

# SHETH L.U.J AND SIR M.V. COLLEGE

SUBJECT NAME: DATA ANALYSIS WITH SPSS/SAS/R

## PRACTICAL 8

**AIM:** Applying basic data cleaning functions: handling missing values using `na.omit()/replace_na()` in R. import dataset.

### >COMMANDS USED

# Load necessary libraries

`library(dplyr)`

`library(tidyr) # Contains replace_na()`

#

=====

=====

# 1. SETUP: Create a Sample Dataset with Missing Values (in R)

#

=====

=====

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**S086**

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# Creating a small data frame with NA values, using column names from your file

# to demonstrate the concepts without needing a separate CSV file.

```
ai_jobs_sample <- data.frame(  
  Job_Title = c("Security Guard", "Research Scientist", "Data  
Scientist", "Software Engineer"),  
  Average_Salary = c(45795, 133355, NA, 136530),  
  Education_Level = c("Master's", NA, "Bachelor's", "PhD"),  
  AI_Exposure_Index = c(NA, 0.62, 0.91, 0.39)  
)
```

```
print("--- 1. Sample Data with NAs (Created in R) ---")
```

```
print(ai_jobs_sample)
```

# Check how many NAs are in each column

```
print("--- Count of Missing Values per Column ---")
```

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```
print(colSums(is.na(ai_jobs_sample)))

# -----
# 2. METHOD A: REMOVE MISSING VALUES (na.omit)
# -----
# Deletes any row that has at least one missing value (NA).

clean_omit <- na.omit(ai_jobs_sample)

print("--- 2. Data after na.omit() ---")
print(paste("Original rows:", nrow(ai_jobs_sample)))
print(paste("Rows remaining:", nrow(clean_omit)))
print(clean_omit)

# -----
# 3. METHOD B: REPLACE MISSING VALUES
(replace_na)
```

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```
# -----  
# This is the "surgical option". We fill missing values with  
logical defaults.  
# Strategy:  
# 1. Average_Salary: Fill with the Mean Salary  
# 2. Education_Level: Fill with "Not Reported"  
# 3. AI_Exposure_Index: Fill missing with 0  
  
# Calculate average salary (ignoring NAs)  
avg_salary <- mean(ai_jobs_sample$Average_Salary, na.rm =  
TRUE)  
  
clean_replace <- ai_jobs_sample %>%  
  replace_na(list(  
    Average_Salary = avg_salary,  
    Education_Level = "Not Reported",  
    AI_Exposure_Index = 0  
  ))
```

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```
print("--- 3. Data after replace_na() ---")
```

```
print(clean_replace)
```

```
# Verify no NAs exist in the columns we cleaned
```

```
print("--- Remaining NAs after replacement (Target columns  
only) ---")
```

```
print(colSums(is.na(clean_replace)))
```

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```
RPROG - RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help

AI Impact on Jobs 2030
Job Title Average Salary Years Experience Education Level AI Exposure Index Tech Growth Factor Automation Probability 2030 Risk C
Showing 1 to 1 of 3,000 entries, 18 total columns

R - R432 --RPROG/
> # Load necessary libraries
> library(dplyr)
> library(tidy) # Contains replace_na()

> # 1. SETUP: Create a Sample Dataset with Missing Values (in R)
> # =====
> # Creating a small data frame with NA values, using column names from your file
> # to demonstrate the concepts without reading a separate CSV file.
+ ai_jobs_sample <- data.frame(
+   Job_Title = c("Security Guard", "Research Scientist", "Data Scientist", "Software Engineer"),
+   Average_Salary = c(45795, 133355, NA, 136530),
+   Education_Level = c("Master's", NA, "Bachelor's", "PHD"),
+   AI_Exposure_Index = c(NA, 0.62, 0.91, 0.39)
+ )
>
> print("---- 1. Sample Data with NAs (Created in R) ----")
[1] "---- 1. Sample Data with NAs (Created in R) ----"
> print(ai_jobs_sample)
  Job_Title Average_Salary Education_Level AI_Exposure_Index
1 Security Guard      45795         Master's              NA
2 Research Scientist 133355           <NA>              0.62
3 Data Scientist    136530      Bachelor's              0.91
4 Software Engineer 136530           PHD                0.39
>
> # Check how many NAs are in each column
> print("---- Count of Missing Values per Column ----")
[1] "---- Count of Missing Values per Column ----"
> print(colSums(is.na(ai_jobs_sample)))
  Job_Title Average_Salary Education_Level AI_Exposure_Index
          0              1              1              1
>
> # 2. METHOD A: REMOVE MISSING VALUES (na.omit)
> # -----
> # Deletes any row that has at least one missing value (NA).
> clean_omit <- na.omit(ai_jobs_sample)
>
> print("---- 2. Data after na.omit() ----")
[1] "---- 2. Data after na.omit() ----"
> print(paste("Original rows:", nrow(ai_jobs_sample)))
[1] "Original rows: 4"
> print(paste("Rows remaining:", nrow(clean_omit)))
[1] "Rows remaining: 3"

Environment History Connections Git Tutorial
R - Global Environment
clean_replace 4 obs. of 4 variables
data_ai_risk 5 obs. of 4 variables
data_base_list 3 obs. of 3 variables
data_job_info 5 obs. of 4 variables
data_main 3000 obs. of 18 variables
data_new_jobs 2 obs. of 3 variables
data_unique_jobs 5 obs. of 18 variables
dropped_multiple 3000 obs. of 16 variables
dropped_one 3000 obs. of 17 variables
dropped_range 3000 obs. of 8 variables
Final_list 5 obs. of 3 variables

Files Plots Packages Help Viewer Presentation
New Folder New File Delete Rename More
Home RPROG
Name Size Modified
.gignore 44 B Dec 1, 2025, 11:03 AM
RData 86.3 KB Nov 24, 2025, 12:31 PM
Rhistory 457 B Dec 1, 2025, 11:12 AM
AI_Impact_on_Jobs_2030.csv 296.4 KB Nov 24, 2025, 11:43 AM
analysis.R
prac.pdfs
README.md 23 B Nov 24, 2025, 6:03 PM
RPROG.Rproj 218 B Dec 1, 2025, 11:12 AM
```

```
RPROG - RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help

AI Impact on Jobs 2030
Job Title Average Salary Years Experience Education Level AI Exposure Index Tech Growth Factor Automation Probability 2030 Risk C
Showing 1 to 1 of 3,000 entries, 18 total columns

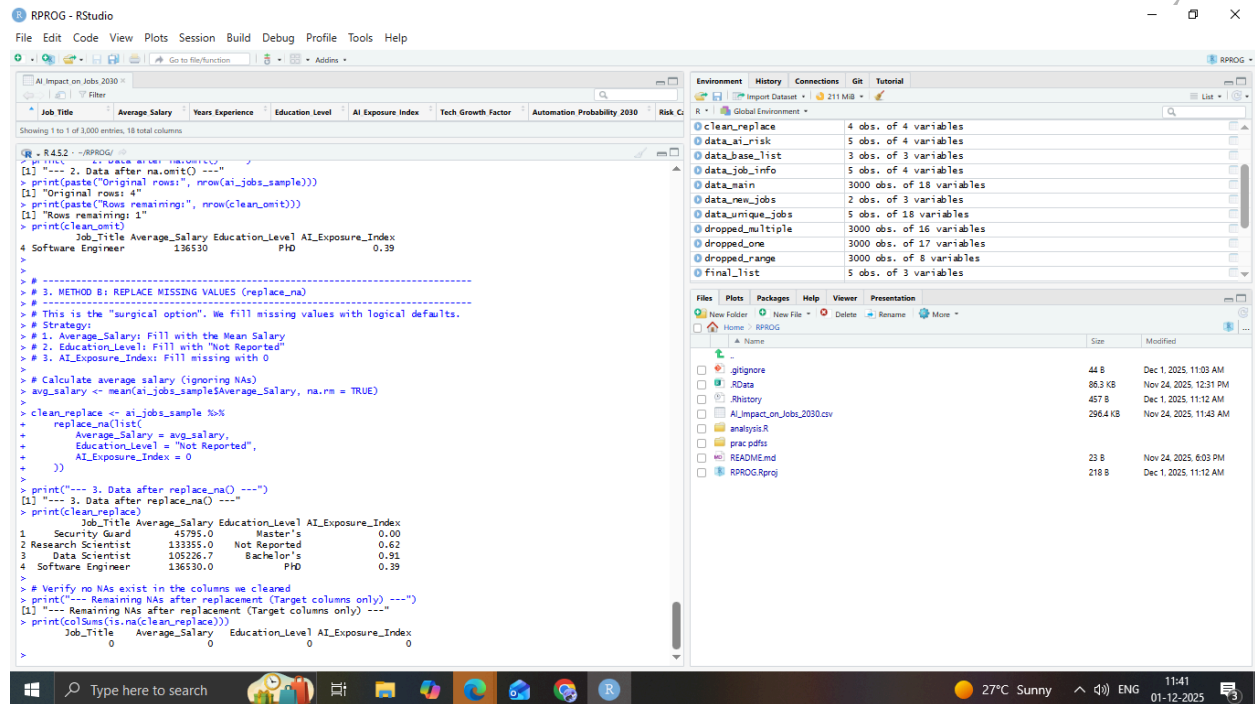
R - R432 --RPROG/
> print("---- 2. Data after na.omit() ----")
[1] "---- 2. Data after na.omit() ----"
> print(paste("Original rows:", nrow(ai_jobs_sample)))
[1] "Original rows: 4"
> print(paste("Rows remaining:", nrow(clean_omit)))
[1] "Rows remaining: 3"
> print(clean_omit)
  Job_Title Average_Salary Education_Level AI_Exposure_Index
4 Software Engineer 136530           PHD                0.39
>
> # 3. METHOD B: REPLACE MISSING VALUES (replace_na)
> # -----
> # This is the "surgical option". We fill missing values with logical defaults.
> # Strategy:
> # 1. Average_Salary: Fill with the Mean Salary
> # 2. Education_Level: Fill with "Not Reported"
> # 3. AI_Exposure_Index: Fill missing with 0
>
> # Calculate average salary (ignoring NAs)
+ avg_salary <- mean(ai_jobs_sample$Average_Salary, na.rm = TRUE)
>
+ clean_replace <- ai_jobs_sample %>%
+   replace_na(list(
+     Average_Salary = avg_salary,
+     Education_Level = "Not Reported",
+     AI_Exposure_Index = 0
+   ))
>
> print("---- 3. Data after replace_na() ----")
[1] "---- 3. Data after replace_na() ----"
> print(clean_replace)
  Job_Title Average_Salary Education_Level AI_Exposure_Index
1 Security Guard      45795.0         Master's              0.00
2 Research Scientist 133355.0      Not Reported              0.62
3 Data Scientist    105265.7      Bachelor's              0.91
4 Software Engineer 136530.0           PHD                0.39
>
> # Verify no NAs exist in the columns we cleaned
> print("---- Remaining NAs after replacement (Target columns only) ----")
[1] "---- Remaining NAs after replacement (Target columns only) ----"
> print(colSums(is.na(clean_replace)))
  Job_Title Average_Salary Education_Level AI_Exposure_Index
          0              0              0              0
```

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The screenshot displays the RStudio environment with the following components:

- Environment Pane:** Lists objects created in the R session:
  - `clean_replace`: 4 obs. of 4 variables
  - `data_ai_risk`: 5 obs. of 4 variables
  - `data_base_list`: 3 obs. of 3 variables
  - `data_job_info`: 5 obs. of 4 variables
  - `data_main`: 3000 obs. of 18 variables
  - `data_new_jobs`: 2 obs. of 3 variables
  - `data_unique_jobs`: 5 obs. of 18 variables
  - `dropped_multiple`: 3000 obs. of 16 variables
  - `dropped_one`: 3000 obs. of 17 variables
  - `dropped_range`: 3000 obs. of 8 variables
  - `final_list`: 5 obs. of 3 variables
- Source Editor:** Contains R code for data cleaning:

```
R - R432 - RPROG.R
# This is the "surge" option. We fill missing values with logical defaults.
# Strategy:
# 1. Average_Salary: Fill with the Mean Salary
# 2. Education_Level: Fill with "Not Reported"
# 3. AI_Exposure_Index: Fill missing with 0

# Calculate average salary (ignoring NAs)
avg_salary <- mean(ai_jobs_sample$Average_Salary, na.rm = TRUE)

clean_replace <- ai_jobs_sample %>%
  replace_na(list(
    Average_Salary = avg_salary,
    Education_Level = "Not Reported",
    AI_Exposure_Index = 0
  ))

print("---- 3. Data after replace_na() ----")
[1] "---- 3. Data after replace_na() ----"
print(clean_replace)
  Job_Title Average_Salary Education_Level AI_Exposure_Index
1 Security Guard 45795.0 Master's 0.00
2 Research Scientist 13355.0 Not Reported 0.62
3 Data Scientist 105226.7 Bachelor's 0.91
4 Software Engineer 136530.0 PhD 0.39

# Verify no NAs exist in the columns we cleaned
print("---- Remaining NAs after replacement (Target columns only) ----")
[1] "---- Remaining NAs after replacement (Target columns only) ----"
print(colSums(is.na(clean_replace)))
  Job_Title Average_Salary Education_Level AI_Exposure_Index
0 0 0 0 0
```
- Files Pane:** Shows the project file structure:
  - `gignore`: 44 B, Dec 1, 2025, 11:03 AM
  - `RData`: 86.3 KB, Nov 24, 2025, 12:31 PM
  - `Rhistory`: 457 B, Dec 1, 2025, 11:12 AM
  - `AI_Impact_on_Jobs_2030.csv`: 296.4 KB, Nov 24, 2025, 11:43 AM
  - `analysis.R`
  - `prac.pdfs`
  - `README.md`: 23 B, Nov 24, 2025, 6:03 PM
  - `RPROG.Rproj`: 218 B, Dec 1, 2025, 11:12 AM

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